DATA ANALYTICS BY DURGA

# Power Query: Data Cleaning & Transformation

## Practical Guide for Three Datasets

**Order Sales • Employee Sales • Banking Transactions**

Purpose: Follow the same Power Query workflow on three progressively more realistic datasets. Do not manually edit the raw source data; perform the cleaning and transformation in Power Query so the process can be refreshed.

## 1. Common Workflow for All Three Datasets

1. Open Excel → Data → Get Data / From Table/Range and bring the required sheet/table into Power Query.
2. Inspect column names, row count, blanks, duplicates, inconsistent text and data types before making changes.
3. Apply transformations in a logical sequence and review Applied Steps after every major operation.
4. Validate the cleaned result before loading it back to Excel.
5. Use Close & Load to create the clean output table.
6. Save the workbook and refresh the query to confirm that the transformation steps are reusable.

## 2. Dataset 1 — Order Sales Practice

Workbook: Power_Query_Data_Cleaning_Practice_Dataset.xlsx

Sheets: Raw_Orders, Feb_Orders, Customers, Products, Cleaned_Example

### A. Data Cleaning Tasks

7. Import Raw_Orders into Power Query.
8. Check the data types of Order ID, Order Date, Quantity, Price and Total Sale.
9. Identify duplicate records. Decide the correct uniqueness rule; Order ID is the intended business identifier for this exercise.
10. Standardize Customer Name by removing unnecessary spaces.
11. Standardize Product values, including variations such as ' gadget '.
12. Identify missing Product, Quantity, Price and Total Sale values. Decide and document how each missing value should be handled.
13. Create or recalculate Total Sale using Quantity × Price.
14. Check whether existing Total Sale values agree with the calculated result.
15. Review the final row count and remaining nulls.

### B. Transformation Tasks

- Create a custom column for Total Sale = [Quantity] * [Price].
- Create a conditional column such as High Sale / Standard Sale based on a threshold chosen by the class.
- Use Group By to calculate total sales by Product.
- Use Group By to calculate total sales by Customer Name.

- Practice Split Column or Merge Columns where appropriate.
- Load the cleaned result to a new worksheet.

### C. Append and Merge Practice

- Append Raw_Orders and Feb_Orders to create a consolidated order table.
- Merge the consolidated orders with Customers using Customer Name or another appropriate available key.
- Merge with Products using Product.
- Try Inner, Left Outer and Left Anti joins and compare the row counts.

## 3. Dataset 2 — Employee Sales Analytics

Workbook: Power_Query_Practice_Dataset_2_Employee_Sales.xlsx

Sheets: Raw_March_April_Sales, May_Sales, Employees, Products

### A. Data Cleaning Tasks

16. Import Raw_March_April_Sales.
17. Set SaleDate as Date, Quantity as Whole Number, UnitPrice and SalesAmount as Decimal Number.
18. Identify and remove the duplicate OrderID.
19. Handle missing Quantity values using a clearly stated business rule.
20. Standardize Region values so North, South, East and West have one consistent format.
21. Check SalesAmount against Quantity × UnitPrice and identify incorrect totals.
22. Standardize Product names.
23. Validate the number of records before and after cleaning.

### B. Transformation Tasks

- Create a calculated SalesAmount column using Quantity × UnitPrice.
- Create a conditional column: High / Medium / Low Sale.
- Use Group By to calculate total sales by Region.
- Use Group By to calculate total sales by Product.
- Create a summary of sales by EmployeeName.
- Extract Year and Month from SaleDate for trend analysis.

### C. Append and Merge Tasks

- Append Raw_March_April_Sales and May_Sales.
- Merge the sales table with Employees using EmployeeName.
- Merge with Products using Product.
- Try Left Outer and Inner joins.
- Use Left Anti Join to find sales records that do not have a matching product record, if any.

## 4. Dataset 3 — Banking Transactions

Workbook: Power_Query_Banking_Practice_Dataset.xlsx

Sheets: June_Transactions, July_Transactions, Customers, Branches

### A. Data Cleaning Tasks

24. Import June_Transactions.
25. Set TransactionDate as Date and Amount as Decimal Number.
26. Identify and remove the duplicate TransactionID.
27. Identify the missing Amount and decide how it should be treated; do not automatically assume that a blank means zero.
28. Standardize BranchCity values such as Mumbai/mumbai and Hyderabad with trailing spaces.
29. Standardize TransactionType values such as deposit/Deposit.
30. Identify the negative amount and investigate whether it is valid or suspicious according to the business rule.
31. Review Status values such as Completed and Pending.
32. Validate row counts and remaining missing values.

### B. Banking Transformations

- Create a conditional column to classify transaction amounts as High, Medium or Low.
- Use Group By to calculate total amount by TransactionType.
- Use Group By to calculate total amount by BranchCity.
- Use Group By to calculate transaction count by AccountType.
- Extract Month from TransactionDate.
- Create a summary of Pending transactions.

### C. Append and Merge Tasks

- Append June_Transactions and July_Transactions.
- Merge transactions with Customers using CustomerID.
- Merge transactions with Branches using BranchCity.
- Try Inner, Left Outer, Right Outer and Full Outer joins.
- Use Left Anti Join to identify transaction records without a matching customer.
- Use Right Anti Join to identify customers without a matching transaction in the selected transaction table.

## 5. Join Types — What Students Must Demonstrate

| Join Type | What to Show | Typical Exercise |
|---|---|---|
| Left Outer | All rows from first table + matching rows from second | All sales/transactions with customer details |
| Right Outer | All rows from second table + matching rows from first | All customers including those without transactions |
| Inner | Only matching records | Transactions with valid customer/product matches |
| Full Outer | All records from both tables | Find matched and unmatched records |
| Left Anti | Rows in first table without a match | Sales/transactions missing a master record |
| Right Anti | Rows in second table without a match | Customers/products with no transaction |

## 6. Final Deliverable for Each Dataset

- One raw/source sheet preserved without manual edits.
- One or more Power Query queries showing the cleaning/transformation steps.
- One clean output table.

- Evidence of at least one duplicate check and missing-value check.
- Evidence of correct data types.
- At least one calculated/custom column.
- At least one Group By summary.
- Append or Merge exercise, as specified for the dataset.
- Short written summary of what was cleaned and transformed.

## 7. Recommended Student Submission

33. Submit the Excel workbook with queries and output tables.
34. Keep query names meaningful, for example: Clean_Orders, Combined_Sales, Clean_Banking.
35. Do not delete the raw source sheets.
36. Take 3–5 screenshots showing important Power Query steps.
37. Write 5 observations from the final cleaned data.
38. Explain one data-quality issue that required a business decision.

## 8. Assessment Rubric — 100 Marks

| Component | Marks |
|---|---|
| Data import and understanding | 10 |
| Data cleaning and data types | 20 |
| Duplicate/missing-value handling | 15 |
| Transformations and calculated columns | 15 |
| Merge/Append/Join practice | 15 |
| Validation and output quality | 10 |
| Analysis/observations | 10 |
| Presentation and documentation | 5 |

## 9. Important Power Query Practice Rules

- Do not overwrite or manually clean the raw source.
- Do not remove duplicate customer names just because the name repeats; determine what defines a duplicate record.
- Do not replace every missing numeric value with zero without a business justification.
- After Merge, check whether the join created unexpected row multiplication.
- After Append, check column names and row counts.
- Always verify totals before and after transformation.
- Use Applied Steps as your audit trail.

## 10. Learning Outcome

By completing all three datasets, students should be able to take messy transaction data, clean and transform it in Excel Power Query, combine related tables, validate the output, and produce an analytics-ready dataset for PivotTables, dashboards and further analysis.